Hormones

Hormones are chemical messengers released into the blood by endocrine glands. They travel to target cells that carry matching receptors, and change what those cells do. This chapter walks through the major glands one by one.

Pancreas

Besides its digestive role, the pancreas contains clusters of endocrine cells called the islets of Langerhans.

Insulin

Insulin is released after a meal when blood glucose rises. It helps muscle, fat and liver cells take up glucose, lowering blood sugar.

Glucagon

Glucagon is released when blood glucose falls. It tells the liver to break down glycogen and release glucose, raising blood sugar.

Beta cells

Beta cells in the islets sense glucose directly and secrete insulin. Alpha cells secrete glucagon. In type 1 diabetes, beta cells are destroyed by the immune system.

Adrenal glands

The two adrenal glands sit on top of the kidneys. Their inner medulla and outer cortex release different hormones.

Adrenaline

Adrenaline (epinephrine) from the medulla drives the fast fight-or-flight response: heart rate rises, airways open and glucose is released within seconds.

Cortisol

Cortisol from the cortex shapes the slower, long-term stress response. It raises blood glucose over hours, suppresses inflammation, and when chronically high it weakens bone.

Thyroid gland

The thyroid in the neck sets the body's metabolic rate with thyroxine (T4) and triiodothyronine (T3).

Calcitonin

Calcitonin, made by the thyroid's C cells, lowers blood calcium by slowing the cells that break down bone.

Parathyroid glands

Four tiny parathyroid glands sit behind the thyroid.

Parathormone

Parathyroid hormone (parathormone) raises blood calcium: it releases calcium from bone, keeps it in the kidneys and activates vitamin D.

Negative feedback

Most hormones are controlled by negative feedback: when a value drifts, a hormone pushes it back, and when it is restored the signal stops — like a thermostat.